CHAPTER ONE

INTRODUCTION

1.1 Introductory Comments

The current interest in neural networks is fueled by the hope that this technology, based on a small model of the human brain, will be able to address classes of problems that are beyond the scope of today's computers. Tasks that are computationally complex for conventional digital computers, such as understanding auditory and visual stimulus, are more efficiently performed by the neural systems of biological organisms. The study of **ANNs (Artificial Neural Networks)** has produced many classes of network architectures in the pursuit of modeling the brain's abilities. A complete description of the various ANNs can be found in Jacek M. Zurada's ***Introduction to Artificial Neural Systems*** [Zur92]. This thesis applies a particular class of such networks developed by Stephen Grossberg et al., under the umbrella of **ART (Adaptive Resonance Theory)**. This theory confronts issues of unsupervised learning, self-organization, stability-plasticity and biological plausibility thus forming a solid foundation on which to construct models of human intelligence.

The Fuzzy ART Map ANN is the latest member of the ART class of networks. Its strengths make it a worthy candidate for further investigation concerning its practical applications. Discussions with various researchers by way of the Internet have revealed some of the applications of the Fuzzy ART Map currently being studied. Daehyon Kim (daehyon.kim@newcastle.ac.uk) is using a Fuzzy ART Map ANN in his research of 3D object recognition for his Ph.D. thesis in Computer Science. Simon Field (s.d.h.field@bnr.co.uk) at the Bell Northern Research neural networks group in Europe is studying Fuzzy ART Map's ability to analyze software complexity metrics in Software Engineering. Gerson Tontini (5mu7tontinig@vmsa.csd.mu.edu) is attempting pattern recognition in quality control charts using Fuzzy ART Map for his Ph.D. thesis in

Mechanical Engineering. Artur Dubrawski (artur.dubrawski@imag.fr) at the Adaptive Systems Lab of the Polish Academy of Sciences in Warsaw has applied Fuzzy ART Map to mobile robot navigation and the strength prediction of high performance concrete. Patrick Wong (patrick@research.petrol.unsw.edu.au) at the Center for Petroleum Engineering at the University of New South Wales in Australia has applied Fuzzy ART Map to rock classification in a petroleum reservoir based on electrical, sonic, and radioactivity measurements. Mehmet Akif Saglam (saglam@bmsr14.usc.edu) at the University of Southern California is applying a Fuzzy ART Map ANN to the function approximation of neurophysiological data collected from different parts of the brain. One can see that the Fuzzy ART Map ANN is being applied to tasks in a variety of disciplines including Computer Science, Geology, and Medical Science, as well as Mechanical, Civil and Software Engineering.

A survey of the literature has turned up no references to the application of the Fuzzy ART Map ANN to the task of **ASR (Automatic Speech Recognition)**. ASR is a vital component of speech understanding systems that promise to improve the ease with which humans can interact with computers. Spoken communication with computers will allow users to remotely access computer based services through standard telephone links, thus making every telephone a potential computer terminal. The demand for this technology is fueled by economic pressures and the rise of mobile cellular technology. As well, many of the world telephones use **PC (Pulse Code)** signaling and hence more sophisticated services that require **DTMF (Dual Tone Multi-Frequency)** signaling are not yet available. It is for these reasons that the development of ASR over telephone channels is imperative. The suitability of the Fuzzy ART Map ANN to the task of ASR is not known. This question could be answered by implementing an ASR system employing a Fuzzy ART Map ANN and testing its performance.

This thesis investigates the applicability of a Fuzzy ART Map ANN to the ASR task. An ASR system employing a Fuzzy ART Map ANN is implemented on a desktop computer using the Matlab programming environment. The audio frequency analysis is

limited to a 300 to 3300 Hz bandwidth so that the system may operate on verbal commands over telephone channels.

1.2 Thesis Outline

This thesis is organized into seven chapters. The first four chapters give a summary of the various fields involved in the subject of this thesis. This material provides the reader with the background information prerequisite to understanding the interdisciplinary discussion that follows in the last three chapters.

Following this introductory chapter, chapter two provides a general summary of neural networks. This chapter discusses the biological inspiration for ANNs by contrasting the manner in which computers and the human brain process information. ANNs are then described at both the individual neuron level and the network level. After a taxonomy of ANN architectures based on learning and recall characteristics, this chapter briefly describes the multi-layer feedforward backpropagation ANN.

Chapter three introduces ART and describes how it is combined with certain aspects of fuzzy logic to produce the Fuzzy ART Map ANN. A discussion of the historical development of the ART model includes a brief description of the competitive learning model from which it evolved. In keeping with its roots in the field of psychology, the mechanics of the ART model are then described with a four stage cognitive process. The merits of ART architectures are presented through a comparison of its learning properties with those of other learning schemes. This chapter then pauses to introduce fuzzy logic, its relationship to neural networks, and the role of the quantified proposition in the Fuzzy ART model. The Fuzzy ART learning process is then presented with an eight stage algorithm. Propositional calculus equations are included along with the text descriptions to cater to readers with a background in formal logic. Finally, the Fuzzy ART Map ANN is presented as a compound or hybrid network consisting of two tightly coupled Fuzzy ART ANNs.

Chapter four highlights the field of ASR including the taxonomy of ASR systems, the stages in the ASR process, and the research areas in which this thesis is involved. The production and perception of speech in humans are then briefly described. The chapter ends

with an overview of **FRT (Formant Ratio Theory)** which attempts to explain how humans perceive speech sounds in terms of the ratios of the resonant frequencies present. This background on FRT is important since speech formant ratios are used in the final ASR system to create temporal pattern vectors which the Fuzzy ART Map ANN component will be trained to recognize.

Chapter five details the implementation of a Fuzzy ART Map ANN as the recognizer component in an ASR system. The chapter is presented within the framework of the four stage ASR process introduced in chapter four. It starts with a description of the creation of the RMC Speech Database, which was used to tune and test the finished system, followed by an account of the recording method and apparatus used to capture the speech signal. Since this system was implemented using a common desktop personal computer, a section on the data storage techniques used to hold the speech database is included. The core of the ASR system is then described in the following three sections which cover the extraction, representation and classification of features present in the speech signal.

Chapter six describes the testing methods used to measure the recognition accuracy of the ASR system. Identical testing is carried out using another ANN in place of the Fuzzy ART Map ANN to provide a performance benchmark. A comparison of the training and testing performance of the two ANNs is discussed. Last, the performance of the Fuzzy ART Map ASR system is tested over a local telephone line. The same training and test sets were used throughout the testing phase to assess the relative effect of the changes made to the ASR system on its recognition accuracy.

Last, chapter seven provides concluding remarks. A summary of the thesis work is presented along with the conclusions reached regarding the use of Fuzzy ART Map ANN in ASR systems. The chapter closes with some suggestions for future work.